



वित्तीय सेवाएं विभाग
DEPARTMENT OF
FINANCIAL SERVICES

PSBs Hackathon Series 2026

(An Initiative of Government of India, Ministry of Finance , Department of Financial Services)



PRESENTS



A i - C S P A R C

AI Powered Innovation for Customer: Security, Privacy, Assurance, Reliance, Convenience

Digital Trust | Ease of doing Business

Promoting Innovation and Fostering Collaboration



SOMAIYA
VIDYAVIHAR UNIVERSITY

K J Somaiya School of Engineering
(formerly K J Somaiya College of Engineering)

PROBLEM STATEMENT & TEAM DETAILS

TEAM NAME (AS SUBMITTED ON PORTAL)		Fantastic Four	
PROBLEM STATEMENT TITLE		Frontline Desk Support in Multilingual Mode Using Gen-AI Voice Assistant	
	NAME	GENDER	AREA OF EXPERTISE
Member 1	CHIRANJIVI PANDA	Male	Machine learning
Member 2	SHIVAM	Male	Machine learning
Member 3	NITIN GUPTA	Male	React, GenAI
Member 4	SHWETA YADAV	Female	Machine learning, GenAI

Gen AI Powered Multilingual Assistant

The Language Barrier Challenge

-  Language barrier create miscommunication in banking interactions
-  Leads to processing delays, transaction errors, and poor customer satisfaction
-  Critical issue in multilingual regions like India (22+ official languages)
-  Causes compliance risks, operational inefficiencies, and lost revenue

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Our Solution

-  Gen-AI powered assistant that acts as a real-time language bridge between customers and bank staff.
-  Captures customer speech and instantly translates it for the staff member.
-  Maintains banking context to ensure accurate interpretation of financial terms.
-  Guides staff with recommended responses and process steps.
-  Generates bilingual summaries of every interaction for records.

Cost-Effective



Uses open-source AI and **BHASHINI** national infrastructure, minimizing implementation and maintenance costs.

Efficient



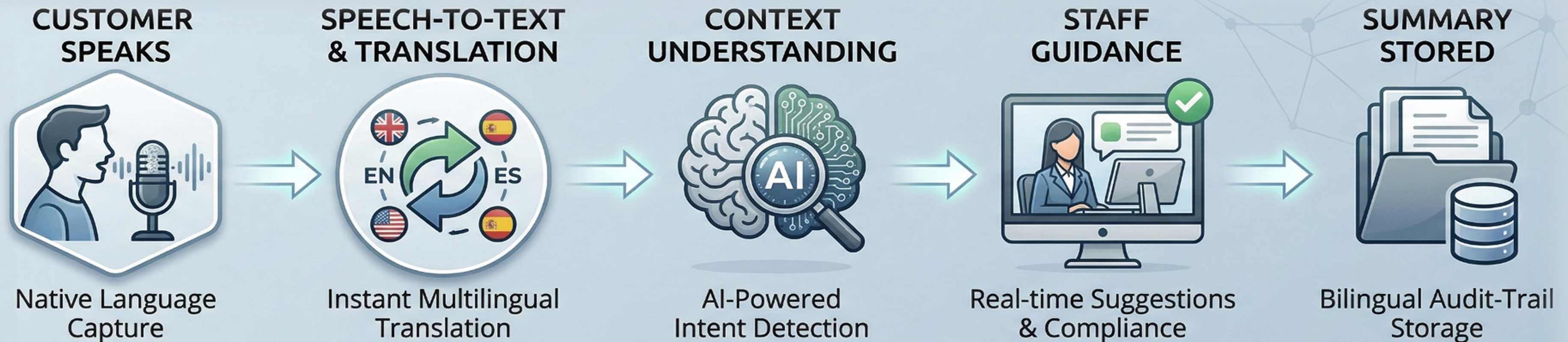
Reduces errors, speeds up service, and improves customer satisfaction.

Scalable



Easily deployable across branches with minimal setup and training.

OUTLINE OF UNIQUE & INNOVATIVE SOLUTION

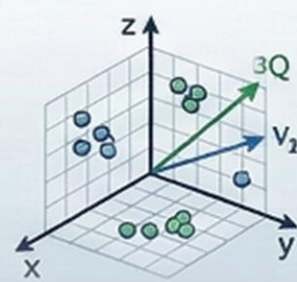


PLATFORM & ARCHITECTURAL FOUNDATIONS



HYBRID AI MODEL

- Open-Source + Proprietary Banking Algs
- Cost-Efficient Performance



LIGHTWEIGHT RAG

- Vector-based Retrieval (pgvector)
- High Contextual Accuracy

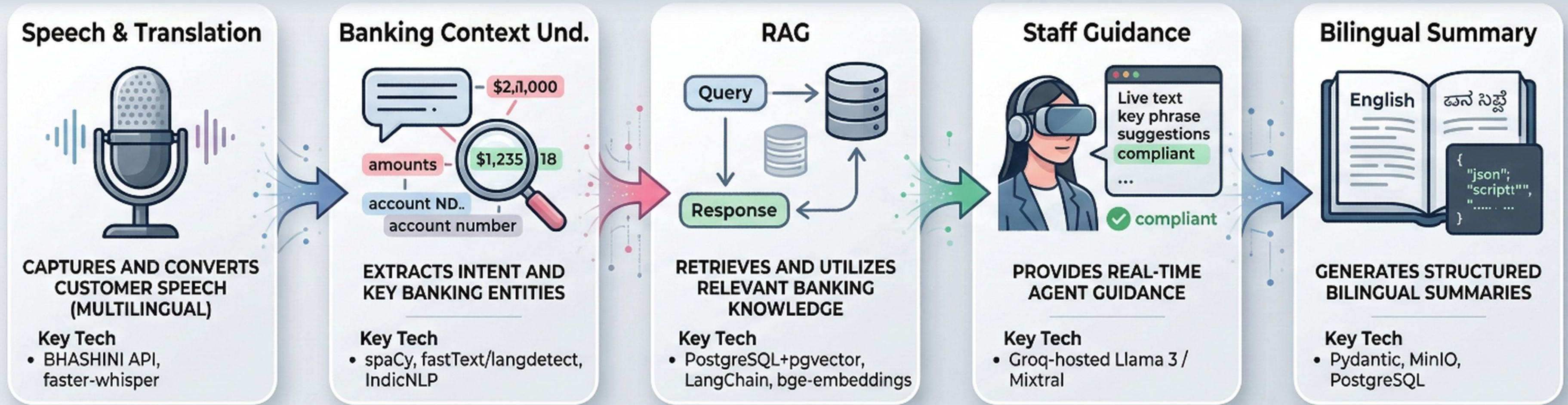


MINIMAL TRAINING

- Prompt Engineering Driven
- Reduced Dev Time & Ops Cost



TECHNICAL APPROACH



Innovative Technical Choices

OPEN-SOURCE AI STACK
(Groq)

HYBRID BHASHINI+OPEN-SOURCE SPEECH

OPTIMIZED LIGHTWEIGHT RAG
(pgvector)

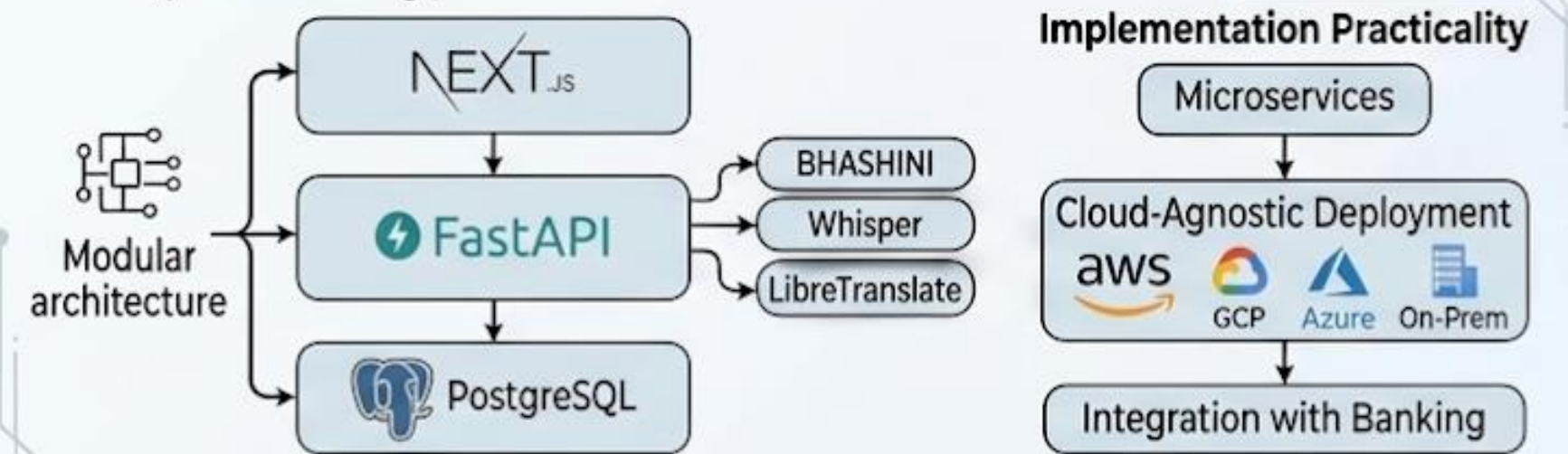
MINIMAL TRAINING WITH PROMPTS

CLOUD-AGNOSTIC DEPLOYMENT

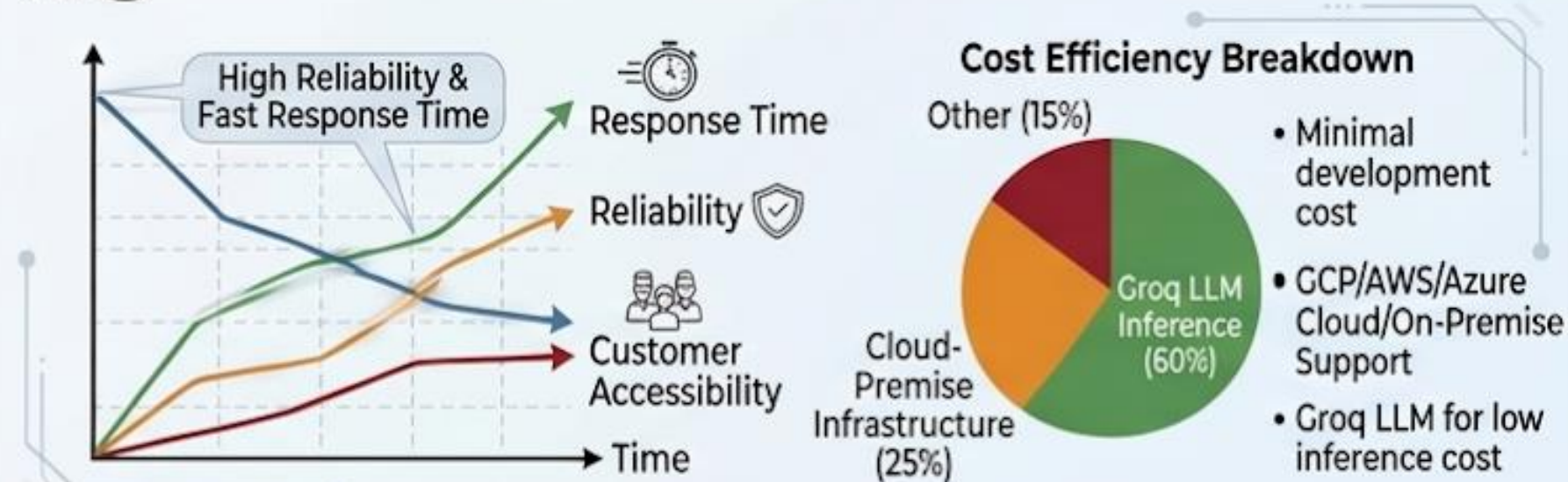


FEASIBILITY & VIABILITY

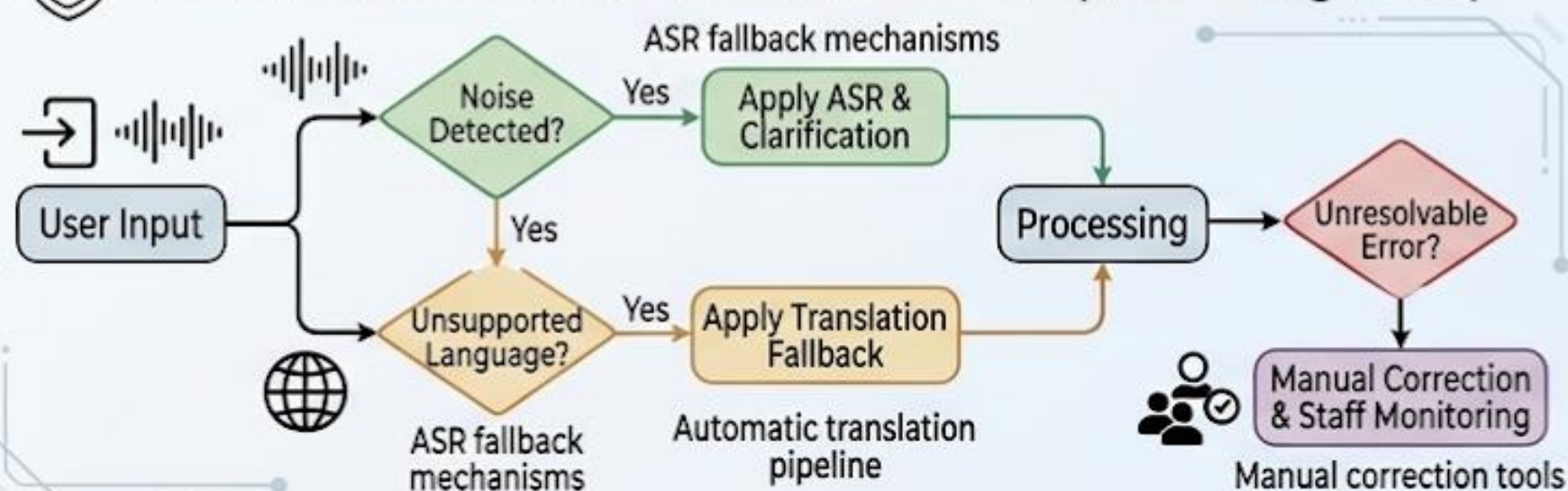
SYSTEM ARCHITECTURE & IMPLEMENTATION FLOW (Feasibility)



PERFORMANCE & COST VIABILITY METRICS (Viability)



RISK HANDLING WORKFLOW (Risk Mitigation)



FEASIBILITY MATRIX

Area	Approach	Outcome
Technology Stack	Open-source tools + APIs	Low development cost
Deployment	Cloud / On-Premise support	Flexible adoption
Reliability	Fallback + validation mechanisms	Stable real-time interaction

IMPACT & BENEFITS



Customer Experience

- Native-language interaction for seamless communication
- Higher satisfaction & trust through clear understanding
- Inclusive banking access for diverse customers



Cost Savings

- Reduced training costs for multilingual staff
- Lower operational errors due to clear communication
- Decreased service overhead through automation



Operational Efficiency

- Faster query resolution and reduced service time
- Less dependency on manual translation
- Higher staff productivity through AI assistance



Compliance & Risk Mitigation

- Regulation-aligned responses for safe interactions
- Structured bilingual documentation for transparency
- Improved audit readiness and traceability



Competitive Advantage

- First-to-market multilingual AI banking solution
- Stronger brand reputation and innovation leadership
- Higher customer retention & acquisition

Operational Cost Reduction with AI Banking Assistant

Open-source AI, BHASHINI, and RAG systems drastically lower operational burn.



ANNUAL SAVINGS

₹2.5 – ₹3.5 Cr



COST REDUCTION

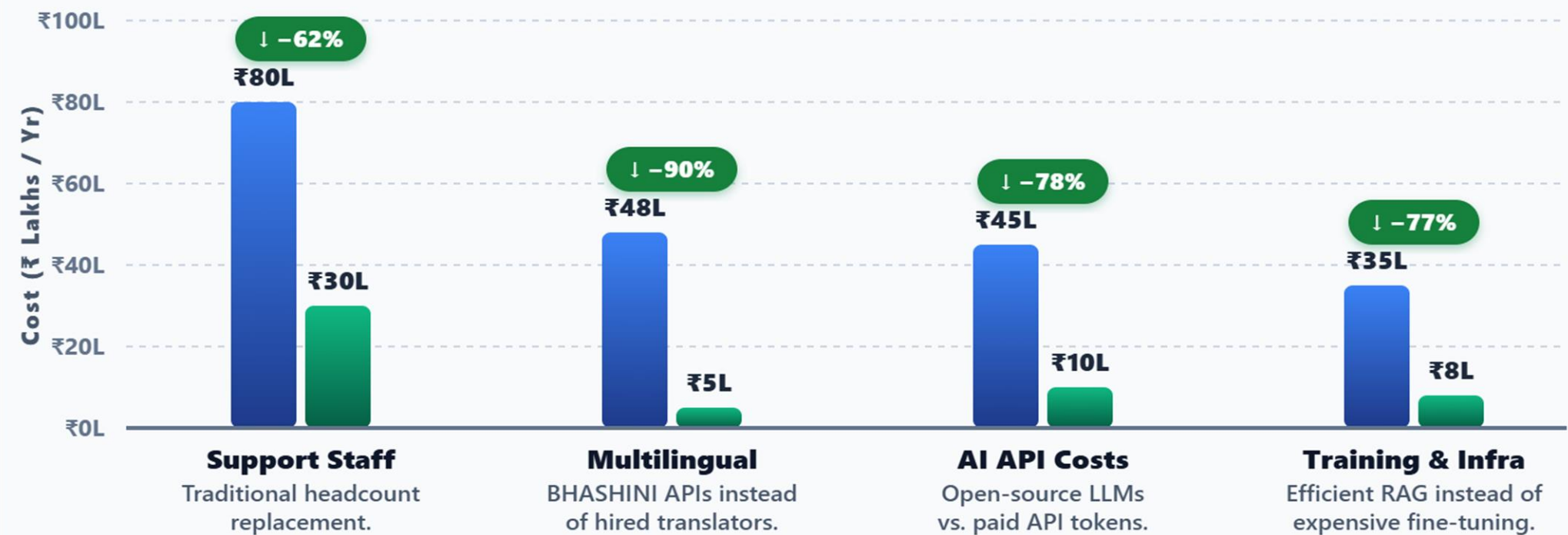
Up to 70–85%



KEY TECH STACK

**OS LLMs + RAG
+ BHASHINI API**

■ Traditional Banking
■ AI-Based Solution



Source: Estimated projections based on banking industry benchmarks and open-source AI infrastructure comparisons.

Impact Summary: Inclusive Banking • Faster Operations • Lower Costs • Stronger Compliance

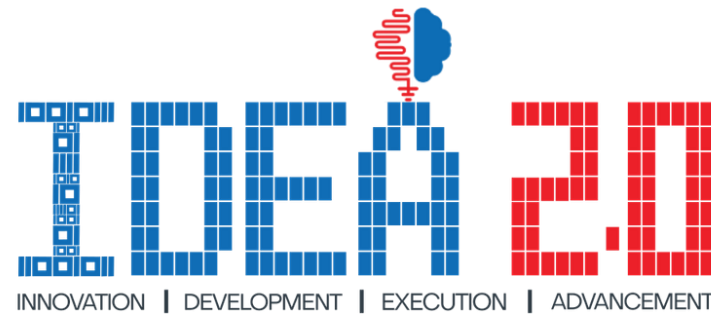


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यूनियन बैंक
ऑफ़ इंडिया
भारत सरकार का उपक्रम



Union Bank
of India
A Government of India Undertaking



BUSINESS MODEL

Scalable SaaS model enabling multilingual banking with recurring revenue, strategic partnerships, and phased market expansion



VALUE CREATION



CUSTOMER SEGMENTS

- Banks & financial institutions
- NBFCs and credit unions
- Future expansion: Government services & telecom



VALUE PROPOSITION

- Multilingual banking access for customers
- Faster and more accurate customer support
- Reduced operational and support costs
- Improved compliance and service efficiency



KEY PARTNERSHIPS

- AI & language technology providers (*BHASHINI, Whisper, Libre Translate*)
- Cloud infrastructure partners for scalable deployment
- Banking institutions for pilot implementations
- Regulatory bodies for compliance and endorsements



REVENUE & GROWTH



REVENUE STREAMS

- SaaS Subscription Model - Tiered pricing based on users, branches, or transactions
- Pay-Per-Use Model - Small fee per interaction or service request
- Enterprise Licencing - Customized solutions for large banks
- Data Analytics Services – Aggregated insights for banks to improve services



GO-TO-MARKET STRATEGY

- Pilot programs with selected banks
- Participation in fintech and banking conferences
- Digital marketing and industry outreach
- Case studies and testimonials from pilot deployments



GROWTH ROADMAP



KEY HIGHLIGHTS



Pilot-Driven Adoption



Recurring Revenue Model



Scalable Across Regions

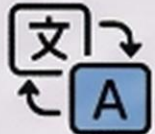
RESEARCH & REFERENCES

Technology Stack & Research Areas



SPEECH RECOGNITION

- Bhashini (Indian Languages)
- Whisper (Fallback)



MACHINE TRANSLATION

- Libre Translate
- Argos Translate (Backup)



RAG INTEGRATION

- pgvector (Vector Search)
- LangChain (Framework)



GENERATIVE AI

- Groq-Hosted LLM (Context & Response)

KNOWLEDGE SOURCES FOR RAG



Banking SOP documents



Product manuals



Compliance guidelines



FAQ documents



RBI circulars

KNOWLEDGE SOURCES



VECTOR DB
(pgvector)

QUERY

RETRIEVAL

GEN AI
(Groq-Hosted LLM)

ANSWER

QUERY

KEY RESEARCH & LINK REFERENCES



Reserve Bank of India

<https://www.rbi.org.in>



NPCI <https://www.npci.org.in>



UIDAI <https://uidai.gov.in>



BHASHINI

<https://bhashini.gov.in>

spaCy – <https://spacy.io>

IndicNLP –
<https://indicnlp.github.io>